3.4 Proofs with Perpendicular Lines

Name _____ Period ____ Date _____

Geometry 3.4 (30-minute plan) · Student sheet

I can…

- find the distance from a point to a line.
- use the Linear Pair Perpendicular Theorem.
- use theorems about lines perpendicular to a transversal.

Vocabulary · fill in the missing word

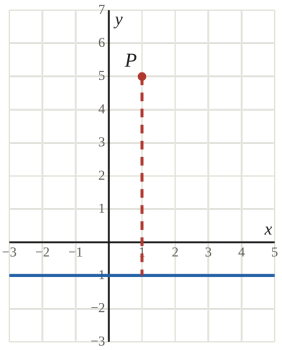
Distance from a point to a line — the length of the _____ segment from the point to the line

Perpendicular lines — two lines that _____ to form a right angle

Part 1 · Notes and examples (fill in each note, then work the example)

Example 1

Find the distance from P(1, 5) to the line y = −1.



Linear Pair Perpendicular Theorem: if two lines intersect to form a linear pair of _____ angles, then the lines are perpendicular.

Example 2

∠1 and ∠2 form a linear pair and ∠1 ≅ ∠2. m∠1 = (4x + 10)°. Find x. Are the lines perpendicular?

Perpendicular Transversal Theorem: in a plane, if a transversal is perpendicular to one of two parallel lines, then it is _____ to the other.

Lines Perpendicular to a Transversal Theorem: in a plane, if two lines are perpendicular to the same line, then they are _____.

Example 3

Lines p and q are both perpendicular to line r (all in one plane). What can you conclude? Why?

Part 2 · Partner practice

Partner practice

In a plane, h ⊥ k and j ⊥ k. Also m ⊥ h. Which lines are parallel? Is m ⊥ j?